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Uncertainty-Aware Hybrid Rendering with Gaussian Splatting and NeRF for High-Fidelity Synthesis

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Introduction

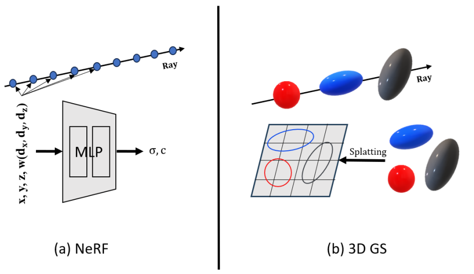

Figure 1. NeRF VS 3D GS

- Real-time rendering in 3D scenes often trades off between speed and quality.
- **3D Gaussian Splatting (3D GS)** is fast but may lack fine detail.
- **NeRF** offers high fidelity but is computationally expensive.
- We propose a **hybrid rendering** pipeline that combines both, guided by **uncertainty estimation**.

Related Work

This project builds upon existing work in NeRF and Gaussian Splatting:

- **NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis** [3] – Introduced NeRF for novel view synthesis but suffers from slow inference.
- **3D Gaussian Splatting for Real-Time Radiance Field Rendering** [2] – Enabled real-time rendering but lacks high-resolution details.
- **SS Mip-NeRF: Supersampled Mip-NeRF** [1] – Supersampling improves Mip-NeRF accuracy with minimal rendering overhead.

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Method

We propose a hybrid rendering framework that leverages the speed of 3D Gaussian Splatting (GS) and the fidelity of NeRF through uncertainty-guided pixel-wise rendering. The full framework consists of three main stages:

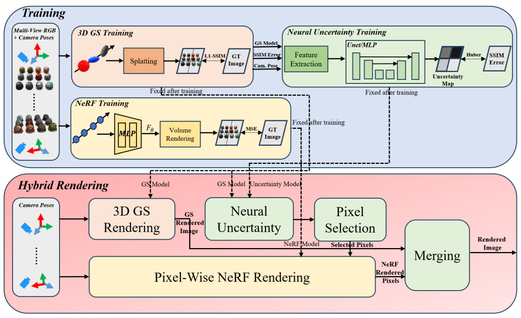

Figure 2. Uncertainty-Aware Hybrid Rendering Framework

1. **3D GS Rendering:** Train a 3D GS model with multi-view RGB and poses, then render views via efficient splatting.
2. **Uncertainty Prediction:** Use a UNet/MLP to predict per-pixel SSIM error from GS features, yielding an uncertainty map.
3. **Pixel-Wise NeRF Rendering:** Re-render top-$k\%$ uncertain pixels with NeRF; merge with GS to form the final output.

Neural Uncertainty Feature

We extract four features from 3D GS: alpha sum, color variance, view direction, and 2D covariance area. These capture visibility, appearance uncertainty, angular sensitivity, and footprint scale, respectively, and are used to predict per-pixel uncertainty.

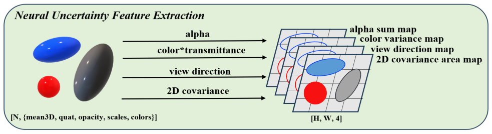

Figure 3. Feature Extraction for Neural Uncertainty Prediction

Qualitative Result

We visualize hybrid rendering results under a 30% uncertainty-based replacement strategy. The predicted uncertainty highlights regions where GS struggles, allowing NeRF to selectively refine them and enhance perceptual quality efficiently.

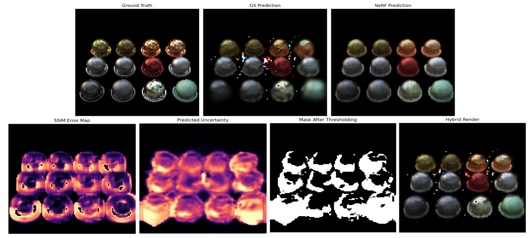

Figure 4. Qualitative results with top 30% highest uncertainty replacement.

Quantitative Result

Although the NeRF Time of the Hybrid model is smaller, most of the time is consumed by GS Time. Experiments show that the 1K-parameter MLP (Spearman: 0.756) performs comparably to the 110K-parameter UNet (Spearman: 0.890), suggesting that smaller networks offer similar hybrid compositing ability with significantly faster processing speed, making it more suitable for this task.

Method	PSNR↑	SSIM↑	GS Time (s)	NeRF Time (s)	Total Time (s)	Replace %
GS only	22.23	0.819	0.210	–	0.210	0%
NeRF only	24.76	0.891	–	0.804	0.804	100%
SS Mip-NeRF	**32.60**	–	–	0.864	0.864	100%
Hybrid-MLP 5%	22.39	0.824	0.766	0.174	0.940	∼5%
Hybrid-MLP 10%	22.64	0.832	0.796	0.221	1.017	∼10%
Hybrid-MLP 30%	24.31	0.875	0.856	0.404	1.260	∼30%
Hybrid-MLP 50%	**24.75**	**0.892**	0.902	0.642	1.544	∼50%
Hybrid-UNet 5%	22.58	0.836	1.148	0.275	1.423	∼5%
Hybrid-UNet 10%	22.88	0.847	1.215	0.357	1.572	∼10%
Hybrid-UNet 30%	24.11	0.880	1.394	0.655	2.049	∼30%
Hybrid-UNet 50%	**24.73**	**0.892**	1.337	0.954	2.291	∼50%

Table 1. Rendering quality and runtime at different replacement rates using MLP vs. UNet for uncertainty prediction.

Note: GS Time for Hybrid refers to the time spent on feature extraction using GS model. NeRF Time for Hybrid refers to the time for ray-marching rendering using NeRF.

References

[1] Jonathan T. Barron, Ben Mildenhall, Matthew Tancik, Peter Hedman, Ricardo Martin-Brualla, and Pratul P. Srinivasan. Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields, 2021.

[2] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3d gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), July 2023.

[3] Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. Nerf: Representing scenes as neural radiance fields for view synthesis. In *ECCV*, 2020.